

JUNSOO HA

San Jose, CA.

🔗 [kuc2477](#) ✦ [in](#) [kuc2477](#) ✦ ✉ junsoo.ha.contact@gmail.com

RESEARCH INTERESTS

Algorithmic Game Theory, Deep Learning , AI Alignment & Safety

EDUCATION

Seoul National University

Master of Science in Artificial Intelligence

Advisor: Prof. Gunhee Kim

Seoul, Korea

Sep. 2020 - Aug. 2023

Hanyang University

Bachelor of Science in Software

Bachelor of Science in Mathematics

Graduated with High Distinction

Seoul, Korea

Mar. 2013 - Aug. 2020

GPA: 4.08 / 4.5

PUBLICATIONS

Stochastic Gradient Descent Ascent is Suboptimal for Nonconvex-PL Min-Max Games

Junsoo Ha

Preprint, 2026

A Horizon-Independent Regret Bound for Optimistic Hedge in General-Sum Games

Junsoo Ha

Preprint, 2026

A Logarithmic Regret Bound for Optimistic Hedge in General-Sum Games

Junsoo Ha

Preprint, 2026

High-Probability Convergence Guarantees of Stochastic Gradient Descent Ascent in Structured Nonconvex Min-Max Games

Junsoo Ha

Conference on Learning Theory (COLT), 2026

On Convergence of Lookahead in Smooth Games

Junsoo Ha, Gunhee Kim

International Conference on Artificial Intelligence and Statistics (AISTATS), 2022

A Neural Dicitale Process Mixture Model for Task-Free Continual Learning

Soochan Lee, **Junsoo Ha**, Dongsu Zhang, Gunhee Kim

International Conference on Learning Representations (ICLR), 2020

Harmonizing Maximum Likelihood with GANs for Multimodal Conditional Generation

Soochan Lee, **Junsoo Ha**, Gunhee Kim

International Conference on Learning Representations (ICLR), 2019

RESEARCH & INDUSTRIAL EXPERIENCE

Unknot Inc.

Quantitative Research Scientist

Built a flexible bespoke research framework that supports any existing financial markets

Built a highly parallelized, error-tolerant production inference system

Seoul, Korea

Aug. 2021 - Aug. 2024

Redesigned a data pipeline from scratch to support thousands of assets across the world
Developed a foundation model that could adapt to multiple financial markets
Raised code quality standards and introduced automated tests to make research reliable

SNU Vision & Learning Lab.

Research Assistant

Seoul, Korea

Nov. 2017 - Aug. 2021

Buzzni

Backend Software Engineer

Seoul, Korea

Mar. 2017 - Jul. 2017

Geopia

Full-Stack Software Engineer

Seoul, Korea

Apr. 2015 - Feb. 2017

Nexol System

Software Engineering Intern

Seoul, Korea

Jan. 2015 - Mar. 2015

LG Electronics

Software Engineering Intern

Seoul, Korea

Jul. 2014 - Aug. 2014

HONORS & AWARDS

Samsung Human-Tech Paper Award

2019

Samsung Electronics

2nd place; Silver Prize

Distinguished Service Award

2019

Hanyang University

RELEVANT COURSEWORKS

Graduate Convex Optimization Theory, Measure Theory

Undergraduate Deep Learning, Mathematical Statistics I-II, Abstract Algebra I-II, Topology I-II, Linear Algebra II, Ordinary Differential Equations, Advanced Calculus I, Vector Calculus, Differential Geometry, Elementary Hyperbolic Geometry

SKILLS

Machine Learning

PyTorch, Tensorflow

Python

Django, Flask, SQLAlchemy, Celery

JavaScript

ES6/7, React, Redux, Electron, Immutable

Database

MySQL, PostgreSQL, SQLite, Redis